### *(i) Lymphocytes*

These are about 20-25% of the total leucocytes. They produce **antibodies** during bacterial and viral infections.

### *(ii) Monocytes*

They are the largest of the leucocytes and are amoeboid in shape. These cells form 5 - 6 % of the total leucocytes.They are **phagocytic** and can **engulf bacteria.**

### Blood Platelets or Thrombocytes



**Thrombocytes**

These are small and colourless. They do not have nucleus. There are about 2,50,000 – 4,00,000 platelets / cubic mm of blood. Life span of platelets is 8–10 days. They play an important role in clotting of blood. Platelets form clot at the site of injury and prevent blood loss.

### More to Know

**Anaemia:** Decrease in number of erythrocytes.
**Leucocytosis:** Increase in the number of leukocytes.
**Leukopenia:** Decrease in number of leukocytes.
**Thrombocytopenia:** Decrease in the number of thrombocytes.

### Functions of blood

i) Transport of respiratory gases (Oxygen and $CO_2$).
ii) Transport of digested food materials to the different body cells.
iii) Transport of hormones.
iv) Transport of nitrogenous excretory products like ammonia, urea and uric acid.
v) It is involved in protection of the body and defense against diseases.
vi) It acts as buffer and also helps in regulation of pH and body temperature.
vii) It maintains proper water balance in the body.

## 14.13 Blood Vessels - Arteries and Veins

Blood vessels are a network of branched tubes that transport blood. There are three types of blood vessels namely **arteries, veins** and **capillaries**

**Arteries:** They are **thick** and **elastic vessels** that carry blood away from the heart to various organs of the body. All arteries carry oxygenated blood except the pulmonary artery which carry deoxygenated blood to the lungs.

**Veins:** Veins are **thin** and **non-elastic vessels** that transport blood to the heart from the different organs. All veins carry deoxygenated blood except the pulmonary vein which carry oxygenated blood from the lungs to the heart.

**Capillaries:** Capillaries are narrow tubes formed by branching of arterioles which then unite to form the venules and veins. They are about 8 μm in diameter. Capillaries are formed of single layer of endothelial cells.

**Table 14.1** Differences between Artery and Vein

| S.No | Artery | Vein |
|---|---|---|
| 1 | Distributing vessel | Collecting vessel |
| 2 | Pink in colour | Red in colour |
| 3 | Deep location | Superficial in location |
| 4 | Blood flow with high pressure | Blood flow with low pressure |
| 5 | Wall of artery is strong, thick and elastic | Wall of vein is weak, thin and non-elastic |
| 6 | All arteries carry oxygenated blood except pulmonary arteries | All veins carry deoxygenated blood except pulmonary veins |
| 7 | Internal valves are absent | Internal valves are present |



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